

SHAMITHA THEERTHA PRASANNA

Bangalore, India, stheerthapra@umass.edu, +91 8296893018, [LinkedIn](#), [Google Scholar](#)

EDUCATION

University of Massachusetts Amherst

M.S. Computer Science (Manning College of Information & Computer Sciences)

B.M.S College of Engineering

B.E. Electronics and Communication Engineering, GPA: 3.4/4.0

Courses: Machine Learning, Deep Learning, Operating Systems, Data Structures, Object-Oriented Programming, Cloud Computing

Incoming Student (Fall 2026)

Amherst, MA

08/2020 - 08/2024

Bangalore, India

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, C++, HTML/CSS

Web & Backend: Flask-compatible frameworks (Spring Boot), Angular, React, Node.js, REST APIs, GraphQL

ML/NLP: Natural language Processing, KeyBERT, TF-IDF, TensorFlow, PyTorch, Scikit-learn, OpenCV

Databases: PostgreSQL, MySQL, MongoDB, Oracle

Other: Git, JIRA, Secure SDLC practices

RESEARCH INTERESTS

NLP, Large Language Models, Multi Modal Learning, Deep Learning, Intersection of Software and AI, AI for Human Behavior

RESEARCH EXPERIENCE

Samsung Research and Development

04/2023 - 11/2023

ML Researcher Intern

Bangalore, India

- Worked on **MultiModal Diarization** which is to identify **Who spoke when?** enhanced with **Who appeared when?**
- Curated a 20+ hour multimodal dataset of **synchronized audio-video recordings** and built a speaker diarization pipeline combining **Resemblyzer**-based audio embeddings with **VGG16 facial embeddings** (via Haar Cascade face detection and OpenCV), clustered using **spectral clustering** to cross-validate speaker identity across modalities.
- Benchmarked the system against four existing diarization approaches (**Whisper/WhisperX/Whisper-JAX + Pyannote**), achieving **7–10% DER on clean audio** vs. 10–15% for standard Whisper+Pyannote pipelines, with consistent gains in overlapping-speech scenarios where audio-only methods degrade sharply.
- Demonstrated that fusing visual cues with audio embeddings improves **diarization robustness** in **multilingual, multi-speaker settings**.

Nokia NBUC

03/2023 - 06/2023

AI Researcher Intern

Bangalore, India (Remote)

- Worked on the problem statement **Developing a module for AutoRobo: An NLP based approach for test validation**, an effort towards automated testing.
- Built an **NLP-driven test-validation pipeline** for AutoRobo that tokenizes and extracts keyphrases from plain-English test-step files, matching them against a structured dataset using fuzzy similarity matching (**FuzzyWuzzy**) to auto-redirect validated steps to output, targeting >80% match accuracy for automated test verification.
- Benchmarked four pre-trained embedding models (**Sentence Transformers, Flair, spaCy, Universal Sentence Encoder**) in conjunction with **KeyBERT** for keyphrase extraction on short-text inputs, evaluated via BERTScore precision/recall/F1.
- Achieved **80% model accuracy** with average BERTScore **precision of 0.84, recall of 0.86, and F1 of 0.85** on the test-validation task which reduced manual effort of 10 hours to an entirely automated process.

INDUSTRY EXPERIENCE

Fidelity Investments

08/2024 – Present

Full Stack Engineer

Bangalore, India

- Develop full-stack database-backed applications using Angular, REST APIs, SQL, and enterprise backend services that support employee lifecycle in the organization.
- Work with **Docker and CI/CD pipelines** to containerize services and automate build/test/deployment workflows.

FSE Intern

Bangalore, India

- Worked towards a solution to automate and digitalize the process of Talent Management by building an application leveraging the **PaaS platform Workday Extend**.
- Involved in extensive Research and Development within Extend exploring new features such as Business Objects, Graph APIs.

PROJECTS

Emotion Detection Using Speech Recognition

11/2022 - 04/2023

- Engineered an end-to-end **speech emotion recognition pipeline** by extracting MFCC features from the RAVDESS dataset and training an **LSTM network** to classify **seven emotion classes** with 86% accuracy.
- Conducted **acoustic and spectral feature analysis** on the DAIC-WOZ depression corpus using MATLAB to investigate speech biomarkers indicative of depressive behavior, enabling data-driven emotion and mental health assessment.
- Evaluated the model through confusion matrix, loss/accuracy curves, and real-time inference on Raspberry Pi 3 audio recordings, demonstrating robust generalization beyond benchmark datasets.

PUBLICATIONS

- [Comparative Analysis of Embedding Models for Keyphrase Extraction: A KeyBERT-Based Approach, IEEE GCAT 2023.](#)
- [Real Time Speech Emotion Recognition using LSTM and Raspberry Pi, IEEE CSITSS 2023.](#)